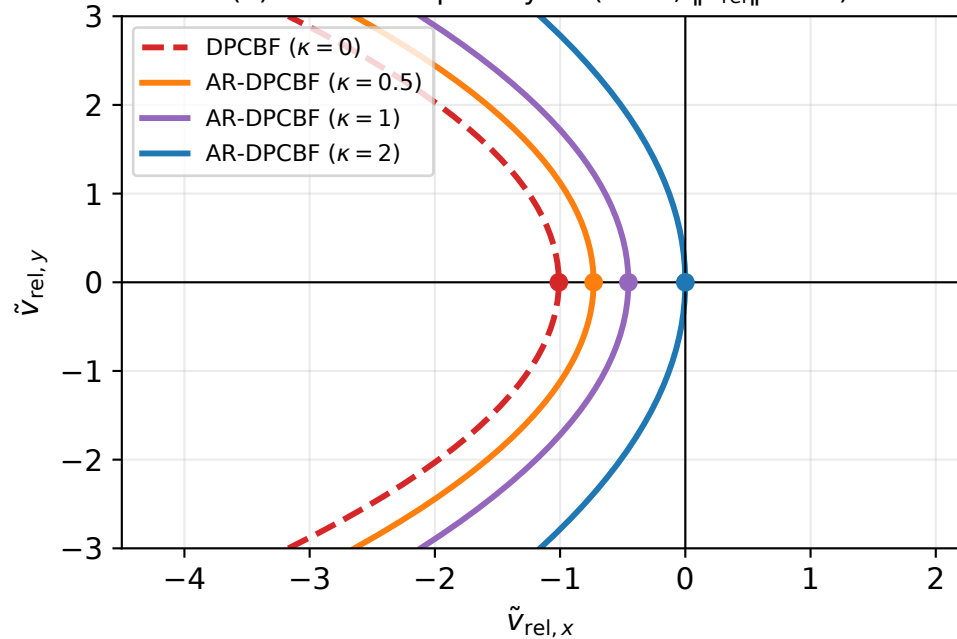
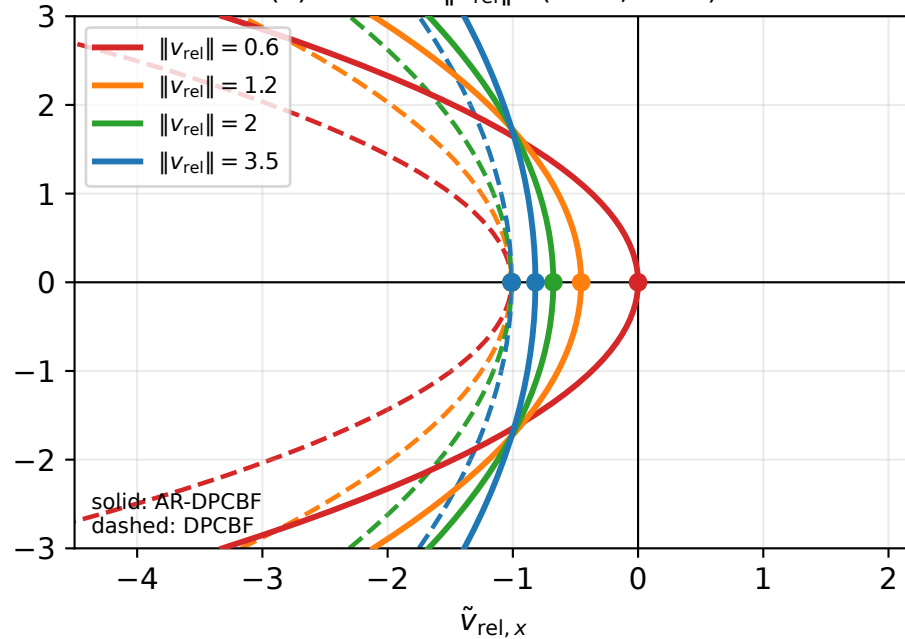


AR-DPCBF safe-set contraction in the LoS velocity frame: the adversarial boundary $\{h^* = 0\}$ (solid) lies to the right of $\{h = 0\}$ (dashed), shrinking the safe set $\{h^* \geq 0\}$

(a) effect of capability κ ($d = 2$, $\|v_{\text{rel}}\| = 1.2$)



(b) effect of $\|v_{\text{rel}}\|$ ($\kappa = 1$, $d = 2$)



(c) effect of clearance d ($\kappa = 1$, $\|v_{\text{rel}}\| = 1.2$)

